2. Requirement Analysis

Introduction

Requirement Analysis is an important phase in system development where the needs and expectations of users are identified and documented. In the **Smart Library Request Workflow in ServiceNow**, this phase focuses on defining the system roles, database structure, and functional capabilities required to implement an efficient library management system.

The goal of this phase is to clearly understand how the system should operate, who will use it, and what features are necessary to support library operations such as requesting books, approving requests, and managing the book catalog.

User Roles

The system is designed with **role-based access control**, which ensures that different users can only perform actions that match their responsibilities. The two primary roles in this system are **Student** and **Librarian**.

Student Role

Students are the main users of the library system. They can access the ServiceNow portal to search for books, check their availability, and submit borrowing requests. Students can also track the status of their requests and view their borrowing history.

Students do not have permission to modify book details or approve requests, ensuring that the system maintains proper control and security.

Example

A student named Ravi logs into the ServiceNow portal and searches for the book " **Data Structures and Algorithms."** The system shows that the book is available. Ravi submits a borrow request, and the request is automatically sent to the librarian for approval.

Librarian Role

The librarian is responsible for managing the library catalog and handling borrow requests. Librarians can add new books, update book details, and track the availability of books in the system.

They also review and approve or reject borrow requests submitted by students. Once a request is approved, the librarian can issue the book and update its status in the system.

Example

A librarian receives Ravi' s borrow request in the ServiceNow workflow dashboard. The librarian checks the book' s availability and approves the request. The system then updates the book status to " **Borrowed"** and records the transaction in the database.

Database Tables

To store and manage information efficiently, the system uses two main database tables: the **Book Table** and the **Borrow Request Table.**

Book Table

The Book Table contains all information related to books available in the library. Each book record includes details such as the book title, author name, category, and availability status.

This table helps the system quickly determine whether a book is available, borrowed, or reserved.

Example Fields

- Book ID
- Book Title
- Author Name
- Category
- Availability Status
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